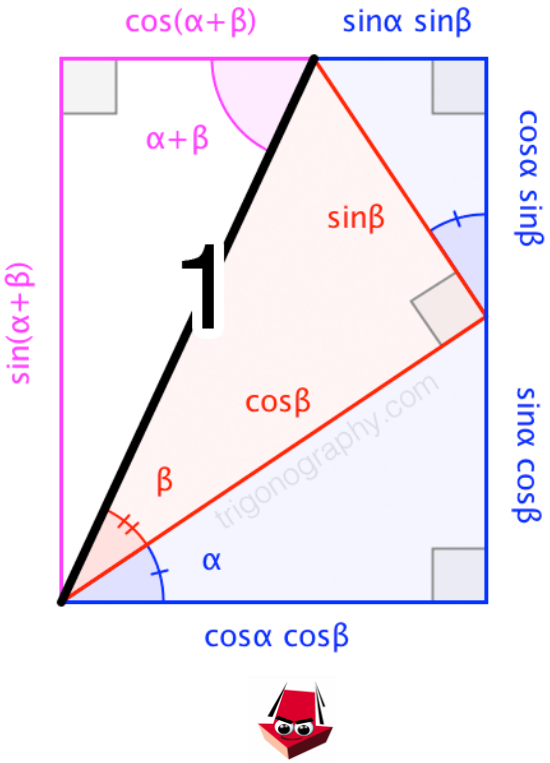
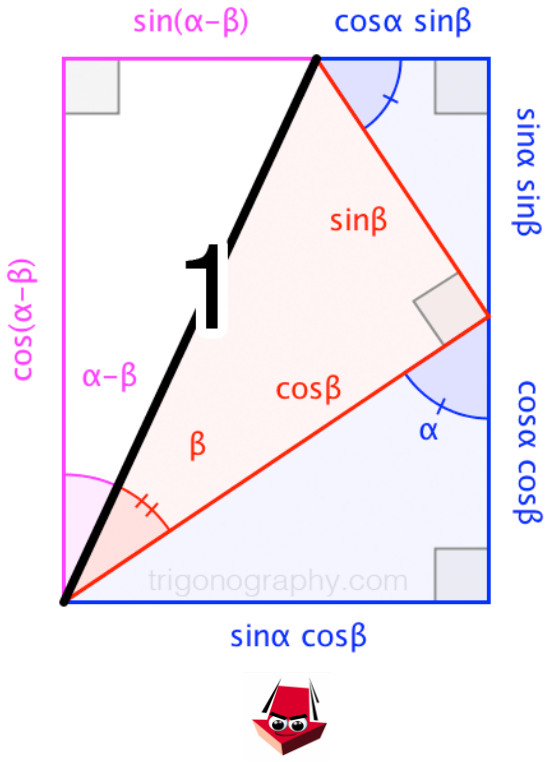
Compound Angles (複角)

In trigonometry, distributive law (分配律) will not work on some cases!

e.g. $\sin(30° + 45°) \neq \sin 30° + \sin 45°$
$\sin(2 \times 30°) \neq 2 \sin 30°$

Therefore, we need to derive Compound Angle Formula to deal with computation of $\sin(\alpha + \beta)$, $\sin(\alpha - \beta)$, $\cos(\alpha + \beta)$, $\cos(\alpha - \beta)$ etc.





$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$
$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

$\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$
$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$

How about $\tan(\alpha + \beta)$ and $\tan(\alpha - \beta)$? Here is the way:

$$\tan(\alpha + \beta) = \frac{\sin(\alpha+\beta)}{\cos(\alpha+\beta)} = \frac{\sin\alpha\cos\beta + \cos\alpha\sin\beta}{\cos\alpha\cos\beta - \sin\alpha\sin\beta} = \frac{\frac{\sin\alpha\cos\beta}{\cos\alpha\cos\beta} + \frac{\cos\alpha\sin\beta}{\cos\alpha\cos\beta}}{\frac{\cos\alpha\cos\beta}{\cos\alpha\cos\beta} - \frac{\sin\alpha\sin\beta}{\cos\alpha\cos\beta}}$$

$$= \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$$

Not for Sale

Compound Angle Formula (複角公式)

- $\sin(A + B) = \sin A \cos B + \cos A \sin B$
- $\sin(A - B) = \sin A \cos B - \cos A \sin B$
- $\cos(A + B) = \cos A \cos B - \sin A \sin B$
- $\cos(A - B) = \cos A \cos B + \sin A \sin B$
- $\tan(A + B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$
- $\tan(A - B) = \dfrac{\tan A - \tan B}{1 + \tan A \tan B}$



註：
測考時試卷會提供複角公式，故同學不需背誦，只需懂得運用便可。

e.g. Find the value of sin 75° in surd form.

$$\begin{aligned}\sin 75° &= \sin(30° + 45°)\\ &= \sin 30° \cos 45° + \cos 30° \sin 45°\\ &= \frac{1}{2}\cdot\frac{\sqrt{2}}{2} + \frac{\sqrt{3}}{2}\cdot\frac{\sqrt{2}}{2}\\ &= \frac{\sqrt{2}+\sqrt{6}}{4}\end{aligned}$$

Let's TRY IT!

Find the value of cos 75° in surd form.

Not for Sale

Compound Angle Formula

[Self-designed Question]

(a) By using $\sin(A + B) \equiv \sin A \cos B + \cos A \sin B$, prove that $\sin 2\theta = 2 \sin\theta \cos\theta$.

(b) By using $\cos(A + B) \equiv \cos A \cos B - \sin A \sin B$, prove that $\cos 2\theta = \cos^2\theta - \sin^2\theta$.

(c) Hence, prove that $\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta}$.

Not for Sale

Double Angle Formulas (倍角公式)

- $\sin 2A = 2\sin A\cos A$
- $\cos 2A = \cos^2 A - \sin^2 A$ **or** $1 - 2\sin^2 A$ **or** $2\cos^2 A - 1$
- $\tan 2A = \dfrac{2\tan A}{1-\tan^2 A}$

- $\sin^2 A = \dfrac{1}{2}(1 - \cos 2A)$
- $\cos^2 A = \dfrac{1}{2}(1 + \cos 2A)$

* $\sin^2 \dfrac{\theta}{2} = \dfrac{1-\cos\theta}{2}$

* $\cos^2 \dfrac{\theta}{2} = \dfrac{1+\cos\theta}{2}$

* 這公式亦稱為半角公式 (Half Angle Formulas)

e.g. Find the value of sin 22.5° in surd form.

Consider $\sin^2 A = \dfrac{1}{2}(1 - \cos 2A)$

$$\sin^2 22.5° = \frac{1-\cos 45°}{2}$$

$$= \frac{1}{2}\left(1-\frac{\sqrt{2}}{2}\right)$$

$$= \frac{2-\sqrt{2}}{4}$$

$$\therefore \quad \sin 22.5° = \frac{\sqrt{2-\sqrt{2}}}{2}$$

注意：
測考時試卷**不會**提供倍角公式(及半角公式)，建議同學緊記相關公式 !!!

Not for Sale

Double Angle Formulas

[題型 (一)：求三角比的準確值 Find the exact value of trigonometric ratios.]

[A.Math 2008 Q.3]

Find the value of $\tan 22.5°$ in surd form. ***(4 marks)***

Not for Sale

Double Angle Formulas

[題型 (二)：證明三角比恒等式 Prove the trigonometric identities.]

[A.Math 1997 Paper II Q.1]

Show that $\dfrac{\sin 3\theta}{\sin \theta} + \dfrac{\cos 3\theta}{\cos \theta} = 4\cos 2\theta$. ***(4 marks)***

Not for Sale

Double Angle Formulas

[題型 (三)：解三角方程 Solve the trigonometric equations.]

[A.Math 1992 Paper II Q.5]

Prove the identity $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$. Hence find the solution of the equation $\sin 2\theta(4\cos^2\theta - 3) - \sin\theta = 0$ for $0 \le \theta \le 2\pi$. ***(6 marks)***

Not for Sale



Product to Sum Formulas (積化和差公式)

The product of two trigonometric ratios can be expressed as sum:

$$\sin(A+B) = \sin A\cos B + \cos A\sin B \quad \ldots\ldots (1)$$
$$\sin(A-B) = \sin A\cos B - \cos A\sin B \quad \ldots\ldots (2)$$



(1) + (2) : $2\sin A\cos B = \sin(A+B) + \sin(A-B)$

(1) – (2) : $2\cos A\sin B = \sin(A+B) - \sin(A-B)$

$$\cos(A+B) = \cos A\cos B - \sin A\sin B \quad \ldots\ldots (3)$$
$$\cos(A-B) = \cos A\cos B + \sin A\sin B \quad \ldots\ldots (4)$$



(3) + (4) : $2\cos A\cos B = \cos(A+B) + \cos(A-B)$

(3) – (4) : $2\sin A\sin B = \cos(A+B) - \cos(A-B)$

Therefore,

- $\sin A\cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$
- $\cos A\sin B = \frac{1}{2}[\sin(A+B) - \sin(A-B)]$
- $\cos A\cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$
- $\sin A\sin B = -\frac{1}{2}[\cos(A+B) - \cos(A-B)]$



註：
測考時試卷會提供積化和差公式，故同學不需背誦，只需懂得運用便可。

將三角比的積 (product) 用和 (sum) 來表示對微積分 (Calculus) 的計算是十分有幫助。

詳情會於中五相關課題介紹。

Not for Sale

Product to Sum Formulas

[題型 (二)：證明三角恒等式 Prove the trigonometric identities.]

[A.Math 2006 Q.2]

Prove the identity $\cos^2 x - \cos^2 y = -\sin(x + y)\sin(x - y)$. ***(3 marks)***

Not for Sale

Product to Sum Formulas

[題型 (三)：解三角方程 Solve the trigonometric equations.]

[A.Math 1990 Paper II Q.5]

Find the solution of the equation $2\sin\frac{x}{2}\sin\frac{3x}{2}=1$ for $0\le x\le 2\pi$. ***(5 marks)***

Not for Sale

Sum to Product Formulas (和差化積公式)

Furthermore, sum of two same trigonometric ratios can be expressed as product:

$$\sin(A+B) + \sin(A-B) = 2\sin A\cos B$$

$$\sin(A+B) - \sin(A-B) = 2\cos A\sin B$$

$$\cos(A+B) + \cos(A-B) = 2\cos A\cos B$$

$$\cos(A+B) - \cos(A-B) = -2\sin A\sin B$$

代 $x = A+B$; $y = A-B$

$\Rightarrow A = \frac{x+y}{2}$, $B = \frac{x-y}{2}$



- $\sin x + \sin y = 2\sin(\frac{x+y}{2})\cos(\frac{x-y}{2})$
- $\sin x - \sin y = 2\cos(\frac{x+y}{2})\sin(\frac{x-y}{2})$
- $\cos x + \cos y = 2\cos(\frac{x+y}{2})\cos(\frac{x-y}{2})$
- $\cos x - \cos y = -2\sin(\frac{x+y}{2})\sin(\frac{x-y}{2})$



註：
測考時試卷會提供和差化積公式，故同學不需背誦，只需懂得運用便可。

和差化積公式 (sum to product formula) 在計算一些三角方程是十分有用的！

e.g. Solve $\cos 3\theta + \cos\theta = 0$ for $0 \le \theta \le 2\pi$.

$$\cos 3\theta + \cos\theta = 0$$

$$2\cos\frac{3\theta+\theta}{2}\cos\frac{3\theta-\theta}{2} = 0$$

$$2\cos 2\theta\cos\theta = 0$$

$\cos 2\theta = 0$ or $\cos\theta = 0$

$2\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$ or $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$

$\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ or $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$

Not for Sale

Sum to Product Formulas (和差化積公式)

[題型 (二)：證明三角恒等式 Prove the trigonometric identities.]

[A.Math 2003 Q.10]

Given two acute angles α and β.

(a) Show that $\dfrac{\sin\alpha+\sin\beta}{\cos\alpha+\cos\beta}=\tan(\dfrac{\alpha+\beta}{2})$.

(b) If $3\sin\alpha-4\cos\alpha=4\cos\beta-3\sin\beta$, find the value of $\tan(\alpha+\beta)$. ***(5 marks)***

Not for Sale

Sum to Product Formulas (和差化積公式)

[題型 (三)：解三角方程 Solve the trigonometric equations.]

[A.Math 1991 Paper II Q.3]

Find the solution of the equation $\cos 4\theta + \cos 2\theta = \cos\theta$ for $0 \le \theta \le 2\pi$. ***(5 marks)***

Not for Sale

HKCEE Questions (level 1)

1. ***[A.Math 1986 Paper II Q.4]***

Find the solution of the equation $\sin 2\theta + \sin 4\theta = \cos\theta$ for $0 \le \theta \le 2\pi$. ***(6 marks)***

Not for Sale

HKCEE Questions (level 1)

2. ***[A.Math 2004 Q.5]***

Find the solution of the equation $\sin 3x + \sin x = \cos x$ for $0 \leq x \leq 2\pi$. ***(5 marks)***

Not for Sale

HKCEE Questions (level I)

3. ***[A.Math 1999 Paper II Q.8]***

(a) Show that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$.

(b) Find the solution of the equation $\cos 6x + 4\cos 2x = 0$ for $0 \le x \le 2\pi$. ***(7 marks)***

Not for Sale

HKCEE Questions (level 2)

4. ***[A.Math 2002 Q.8]***

Given $0 < x < \frac{\pi}{2}$. Show that $\frac{\tan x - \sin^2 x}{\tan x + \sin^2 x} = \frac{4}{2 + \sin 2x} - 1$.

Hence, or otherwise, find the least value of $\frac{\tan x - \sin^2 x}{\tan x + \sin^2 x}$. ***(5 marks)***

Not for Sale

HKDSE Questions (Level 2)

5. ***[M2 2013 Q.7]***

(a) Prove the identity $\tan x = \dfrac{\sin 2x}{1+\cos 2x}$.

(b) Using (a), prove the identity $\tan y = \dfrac{\sin 8y \cos 4y \cos 2y}{(1+\cos 8y)(1+\cos 4y)(1+\cos 2y)}$.

(7 marks)

Not for Sale

HKDSE Questions (level 2)

6. ***[M2 2020 Q.3]***

(a) Let x be an angle which is not a multiple of 30°. Prove that

(i) $\tan 3x = \dfrac{3\tan x - \tan^3 x}{1 - 3\tan^2 x}$

(ii) $\tan x \tan(60° - x)\tan(60° + x) = \tan 3x$

(b) Using (a)(ii), prove that $\tan 55° \tan 65° \tan 75° = \tan 85°$.

(6 marks)

Not for Sale

HKDSE Questions (Level 3)

7. ***[M2 2015 Q.7]***

(a) Prove that $\sin^2 x \cos^2 x = \dfrac{1-\cos 4x}{8}$.

(b) Let f(x) = $\sin^4 x + \cos^4 x$

(i) Express f(x) in the form $A\cos Bx + C$, where A, B and C are constants.

(ii) Solve the equation 8f(x) = 7 for $0 \le x \le \dfrac{\pi}{2}$.

(7 marks)

Not for Sale

HKDSE Questions (Level 3)

8. ***[M2 2018 Q.3]***

(a) If $\cot A = 3\cot B$, prove that $\sin(A+B) = 2\sin(B-A)$.

(b) Using (a), solve the equation $\cot(x+\frac{4\pi}{9}) = 3\cot(x+\frac{5\pi}{18})$, where $0 \le x \le \frac{\pi}{2}$.

(5 marks)

Not for Sale

HKDSE Questions (Level 3)

9. ***[M2 2021 Q.4]***

(a) Prove that $\cos 2x + \cos 4x + \cos 6x = 4\cos x \cos 2x \cos 3x - 1$.

(b) Solve the equation $\cos 4\theta + \cos 8\theta + \cos 12\theta = -1$, where $0 \le \theta \le \frac{\pi}{2}$.

(6 marks)

Not for Sale

HKDSE Questions (Level 3)

10. ***[M2 2017 Q.7]***

(a) Prove that $\sin 3x = 3\sin x - 4\sin^3 x$.

(b) Let f(x) = $\frac{\pi}{4} < x < \frac{\pi}{2}$.

(i) Prove that $\dfrac{\sin 3\left(x - \frac{\pi}{4}\right)}{\sin\left(x - \frac{\pi}{4}\right)} = \dfrac{\cos 3x + \sin 3x}{\cos x - \sin x}$.

(ii) Solve the equation $\dfrac{\cos 3x + \sin 3x}{\cos x - \sin x} = 2$. ***(8 marks)***

Not for Sale

HKCEE Questions (Level 4)

11. ***[A.Math 1988 Paper II Q.7]***

(a) Without using calculators, show that $\frac{\pi}{10}$ is a root of $\cos 3\theta = \sin 2\theta$.

(b) Given that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ and $\sin 2\theta = 2\sin\theta\cos\theta$, find the value of $\sin\frac{\pi}{10}$, expressing the answer in surd form.

(7 marks)

Not for Sale

HKDSE Questions (Level 4)

12. ***[M2 2016 Q.6]***

(a) Prove that $x + 1$ is a factor of $4x^3 + 2x^2 - 3x - 1$.

(b) Express $\cos 3\theta$ in terms of $\cos\theta$.

(c) Using the results of (a) and (b), prove that $\cos\frac{3\pi}{5} = \frac{1-\sqrt{5}}{4}$. ***(6 marks)***

Not for Sale

HKDSE Questions (level 4)

13. ***[M2 2015 Q.8]***

(a) Using mathematical induction, prove that $\sin\frac{x}{2}\sum_{k=1}^{n}\cos kx = \sin\frac{nx}{2}\cos\frac{(n+1)x}{2}$

for all positive integers n.

(b) Using (a), evaluate $\sum_{k=1}^{567}\cos\frac{k\pi}{7}$. ***(8 marks)***

Not for Sale

HKDSE Questions (level 4)

Not for Sale

HKCEE Questions (level 5)

14. ***[A.Math 1995 Paper II Q.9]***

(a) Show that $\cos^2 A - \cos^2 B = \sin(A + B)\sin(B - A)$. ***(3 marks)***

(b) ABC is a triangle,

(i) Using (a), show that

$$\cos^2 A - \cos^2 B + \sin^2 C = 2\cos A \sin B \sin C.$$

(ii) If $\cos^2 A - \cos^2 B - \cos^2 C = -1$, show that ABC is a right-angled triangle.

(6 marks)

(c) Using (a), or otherwise, show that

$$\cos^2 x - \sin^2 y = \cos(x + y)\cos(x - y).$$

Hence find the solution of

$$\cos^2 2\theta - \sin^2 3\theta + \cos 2\theta \sin 5\theta = 0 \text{ for } 0 \le \theta \le 2\pi.$$ ***(7 marks)***

Not for Sale

HKCEE Questions (level 5)

Not for Sale

HKCEE Questions (level 5)

15. ***[A.Math 1991 Paper II Q.8]***

(a) Given that $\tan x = k \tan y$

(i) Show that $\sin(x + y) = (k + 1)\cos x \sin y$.

(ii) Hence show that

$$(k + 1)\sin(x - y) = (k - 1)\sin(x + y).$$ ***(6 marks)***

(b) Gven that $\tan(\theta + 10°) = k \tan(\theta - 20°)$ has solution

(i) Show that $\sin(2\theta - 10°) = \dfrac{k+1}{2(k-1)}$.

(ii) Hence find the range of possible values of k.

(7 marks)

(c) Find the solution of

$$\tan(\theta + 10°) = -2 \tan(\theta - 20°),$$

giving the answer correct to the nearest 0.1 degree. ***(3 marks)***

Not for Sale

HKCEE Questions (level 5)

Not for Sale

HKCEE Questions (level ∞)

16. ***[A.Math 1997 Paper I Q.11 Modified]***

(a) Solve the equation $\cos 5\theta = 0$ for $0° \le \theta \le 180°$. ***(2 marks)***

(b) It is given that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ and $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$, show that $\cos 5\theta = 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$. ***(4 marks)***

(c) Let $f(x) = 16x^4 - 20x^2 + 5$

(i) By putting $x = \cos\theta$ and using the result of (a) and (b), find the four values of θ for which $\cos\theta$ is a root of $f(x) = 0$.

Hence show that

$$f(x) = 16(x^2 - \cos^2 18°)(x^2 - \cos^2 54°) \quad \ldots\ldots \quad (*)$$

(ii) Using (*), form a quadratic equation with integral coefficients whose roots are $\sin^2 18°$ and $\sin^2 54°$.

[Hints: You may treat $f(x)$ as a quadratic function of x^2.] ***(10 marks)***

Not for Sale

HKCEE Questions (level ∞)

Not for Sale

More Questions in Past Papers

1. ***[A.Math 1982 Paper II Q.3]***

Find the solution of the equation $\cos 2\theta - \sqrt{3}\cos\theta + 1 = 0$ for $0 \le \theta \le 2\pi$. ***(6 marks)***

2. ***[A.Math 1980 Paper I Q.7]***

Find the solution of the equation $\tan 7\theta + \cot 2\theta = 0$ for $0 \le \theta \le 2\pi$. ***(6 marks)***

3. ***[A.Math 1983 Paper II Q.7]***

Show that $\sin^2 n\theta - \sin^2 m\theta = \sin(n+m)\theta \sin(n-m)\theta$.

Hence, or otherwise, solve the equation $\sin^2 3\theta - \sin^2 2\theta - \sin\theta = 0$ for $0 \le \theta \le \pi$.

(7 marks)

4. ***[A.Math 1998 Paper II Q.7]***

Show that $\sin(3x + \frac{\pi}{4})\cos(3x - \frac{\pi}{4}) = \frac{1+\sin 6x}{2}$.

Hence, or otherwise, find the solution of the equation $\sin(3x + \frac{\pi}{4})\cos(3x - \frac{\pi}{4}) = \frac{3}{4}$ for $0 \le x \le 2\pi$.

(6 marks)

5. ***[A.Math 1987 Paper II Q.6]***

Express $\sin 3\theta$ in terms of $\sin\theta$.

Hence find the three roots of the equation $8x^3 - 6x + 1 = 0$ to 2 significant figures. ***(6 marks)***

Title of the publication:	Compound Angles
Copyright owner:	HKEAA
Date of making this copy:	1 – 9 – 2023
Some material in this note has been copied under a license issued by the HKEAA. You are not permitted to make any further copy of this work, or to make It available to others. It is important to understand and respect copyright.	

Not for Sale